Witch flounder (*Glyptocephalus cynoglossus*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	2.44	1.5	
Complexity in Reproductive Strategy	1.68	2.55	
Dispersal of Early Life Stages	1.2	2.7	
Egg and Larval Survival and Settlement Requirements	1.6	1.8	
Historical Variability in Temperature	2.12	3	
Juvenile Habitat Requirements	1.12	3	
Other Stressors	1.6	2.05	
Population Growth Rate	3.52	2.58	
Prey Specificity	2.64	2.9	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.36	2.7	
Spawning Cycle	1.8	2.7	
Stock Size/Status	3.92	2.3	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.23	1.07	
Bottom Oxygen	2.67	1.02	
Bottom Salinity	2.68	1.06	
Bottom Aragonite Solubility	1.38	0.84	
Net primary production	1.04	0.10	
Downward Flux of Particulate Organic Carbon	1.32	0.34	
Total Exposure	**Moderate (1.98)**	**0.93**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (5.93)**	**2.8**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Witch flounder. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	High	None
Exposure	High	Moderate	Decrease
Vulnerability	High	Moderate	Decrease
Directional Effect	Negative	Negative	None

Stocks

Witch flounder is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

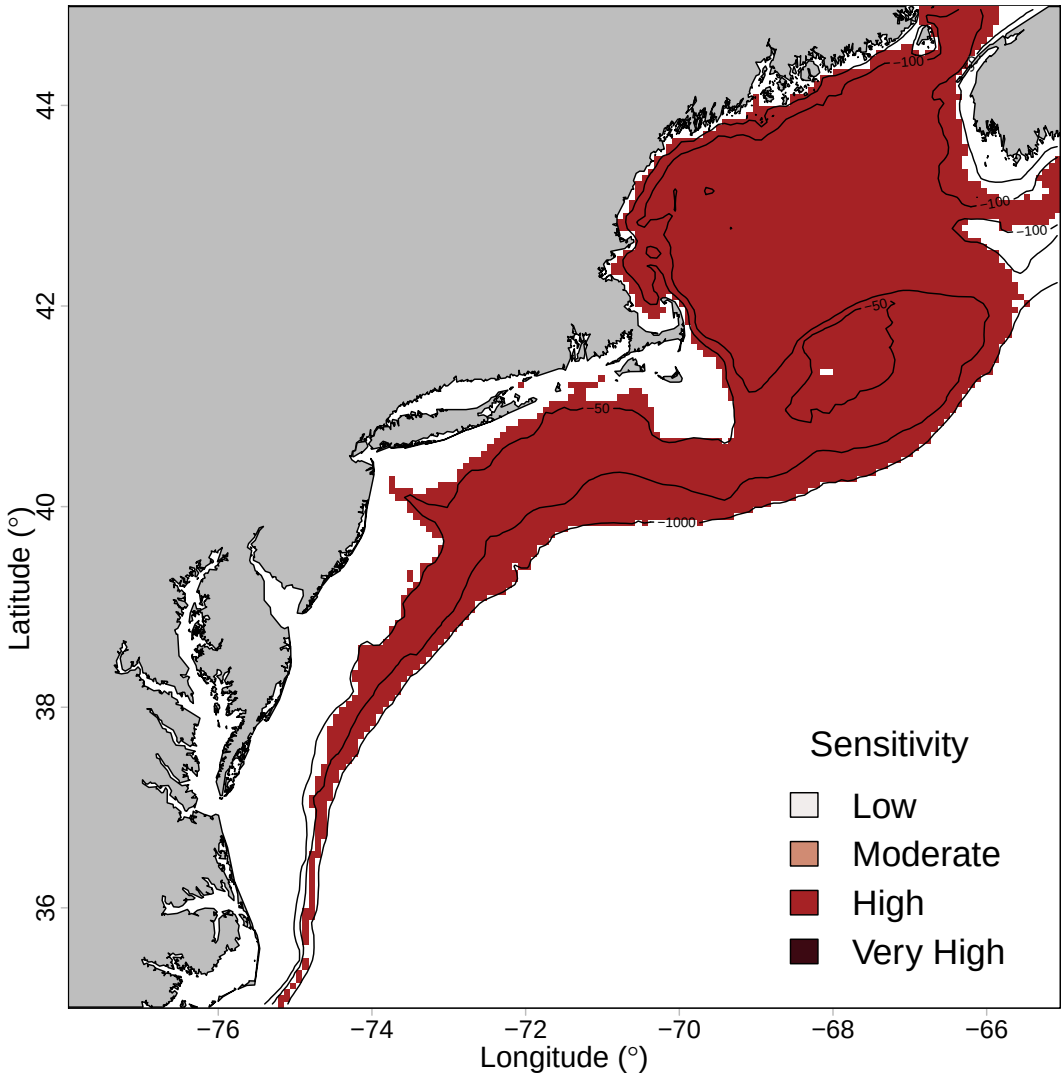


Figure 2: Witch flounder Sensitivity on the Northeast Shelf

Witch flounder received a **High** sensitivity score, with 100% certainty. Two attributes had scores greater than 3: Stock Size (3.9) and Population Growth Rate (3.5) (Figure 1).

Winter flounder are a long-lived, slowly maturing species, reaching maturity at 4.5-6 years and living up to 30 year (Burnett, Ross, and Clark 1992) (Cargnelli et al. 1999). The stock status is poorly understood; the stock is overfished but it is unclear if overfishing is occurring due to lack of data (Northeast Fisheries Science Center (NEFSC) 2024).

Witch flounder also received a High sensitivity score in the 2016 Climate Vulnerability Assessment. This was also driven by Stock Size and Population Growth Rate (Hare et al. 2016).

Exposure:

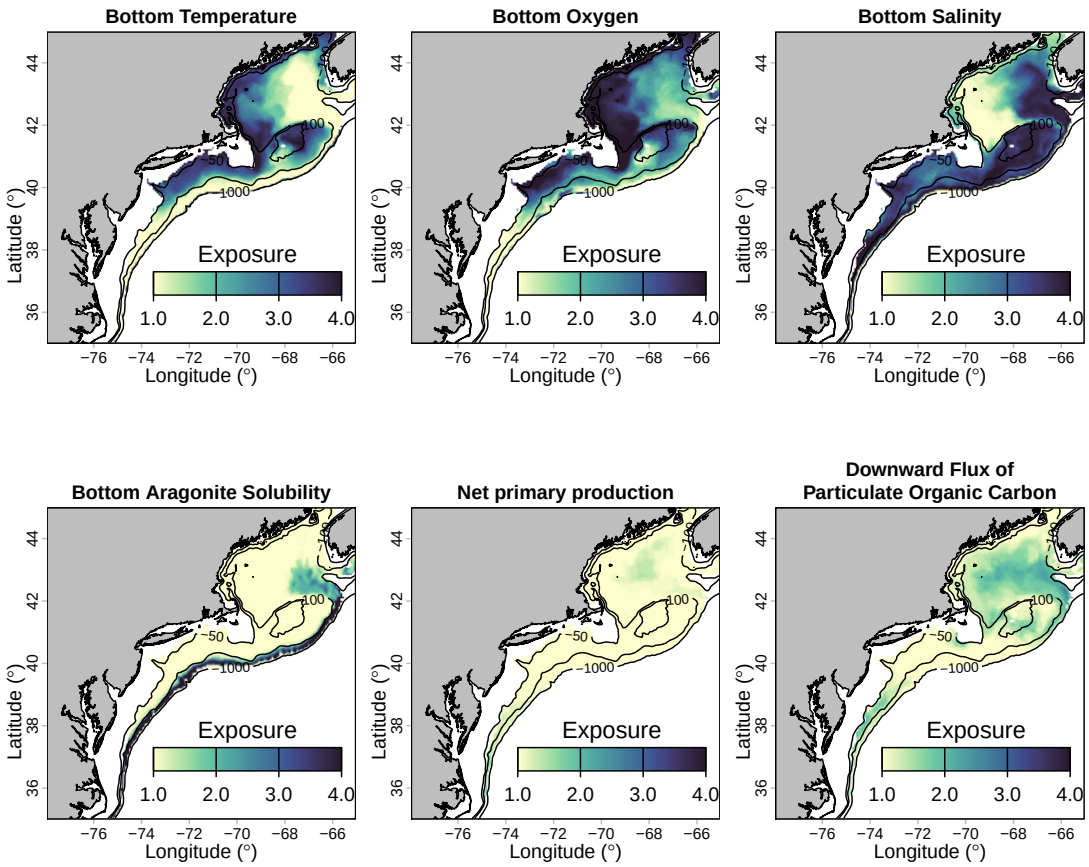


Figure 3: Maps of each Exposure Factor for Witch flounder. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Witch flounder was **Moderate** (1.98 +/- 0.93) (Figure 1). Exposure was Very High over Georges Bank and High in southern New England, along the coast of the Gulf of Maine, and along the continental shelf break (Figure 4). These patterns were driven by bottom temperature, oxygen, and aragonite solubility exposures (Figure 3).

In the previous Assessment, witch flounder received a High Exposure score, driven by sea surface temperature and ocean acidification (Hare et al. 2016).

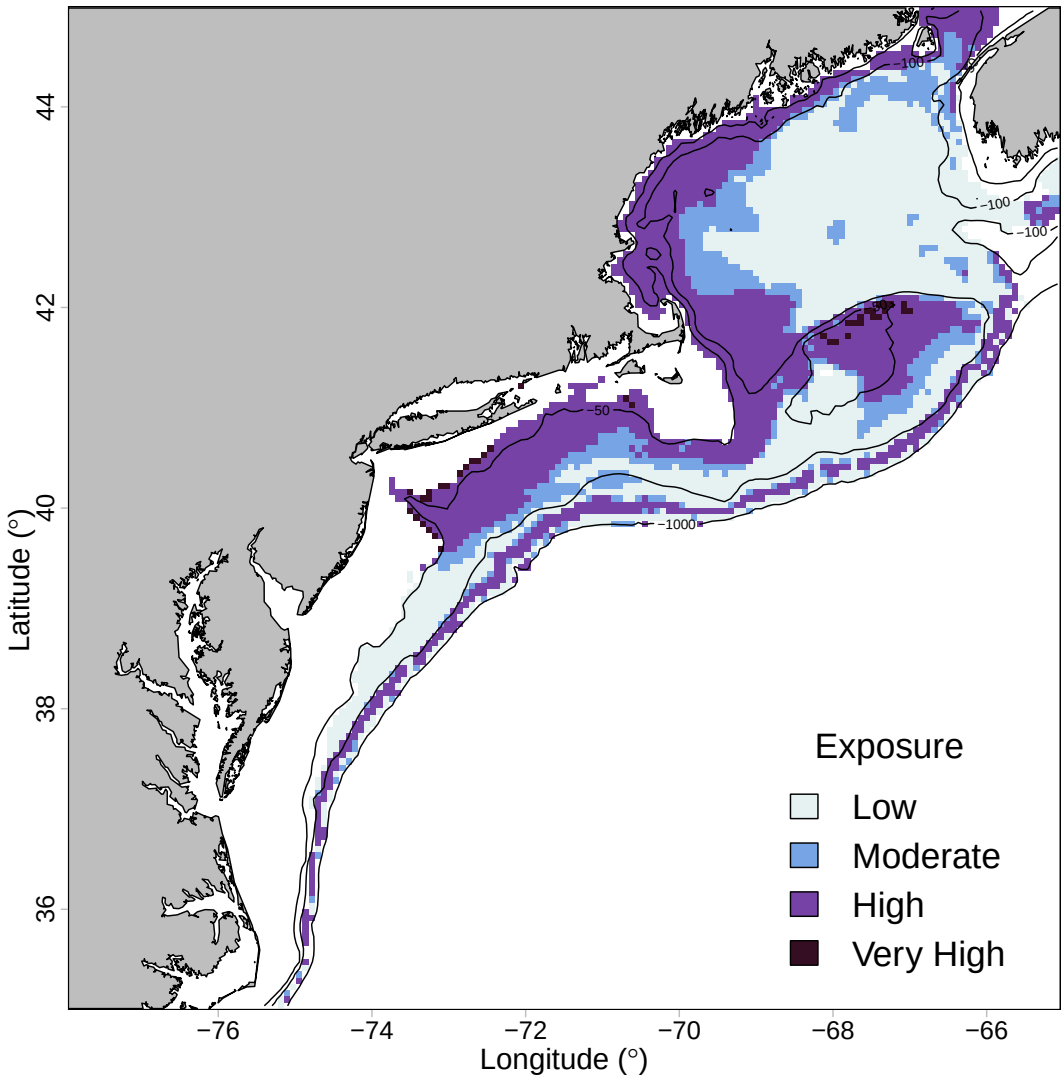


Figure 4: Total Exposure on the Northeast Shelf for Witch flounder. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

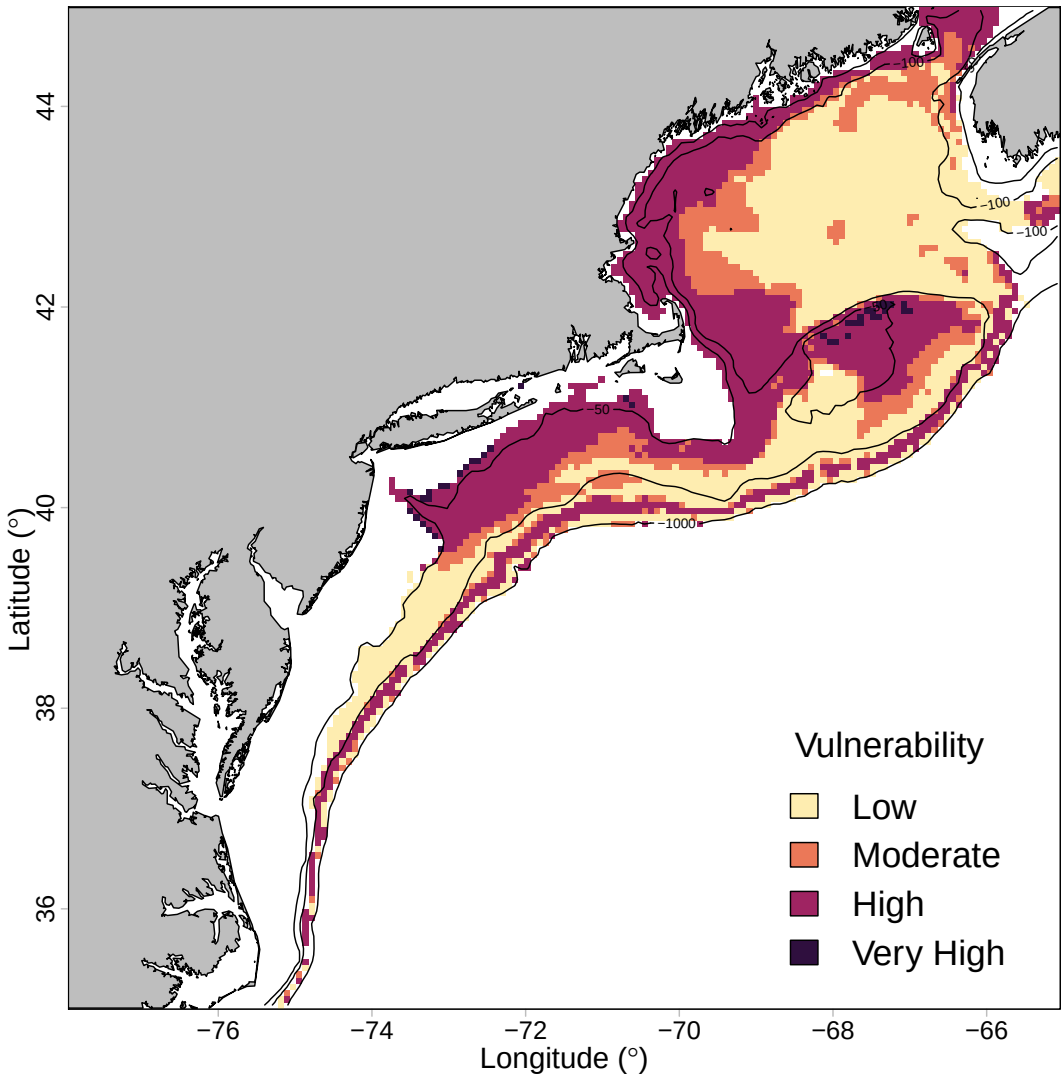


Figure 5: Vulnerability on the Northeast Shelf for Witch flounder. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Witch flounder was **Moderate** (5.93 +/- 2.80) (Figure 1). Vulnerability was Very High over Georges Bank and High in Southern New England, along the coast of the Gulf of Maine, and on the shelf break (Figure 5).

In the previous assessment, Witch flounder received a High Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

78% of data quality scores were greater than or equal to 2, indicating good data quality. Both Adult Mobility and Egg and Larval Survival and Settlement Requirements were identified as data gaps (mean

attribute data quality score < 2) (Figure 1).

Model Confidence Score:

Mean model confidence was 2.1 (+/- 0.4) indicating moderate confidence in the models. Experts stated that predicted distributions met expectations, but the model tended to under-predict species presence in the Gulf of Maine. The ensemble AUC score was 0.91, indicating excellent prediction accuracy.

Additional Metrics

Distribution Shifts:

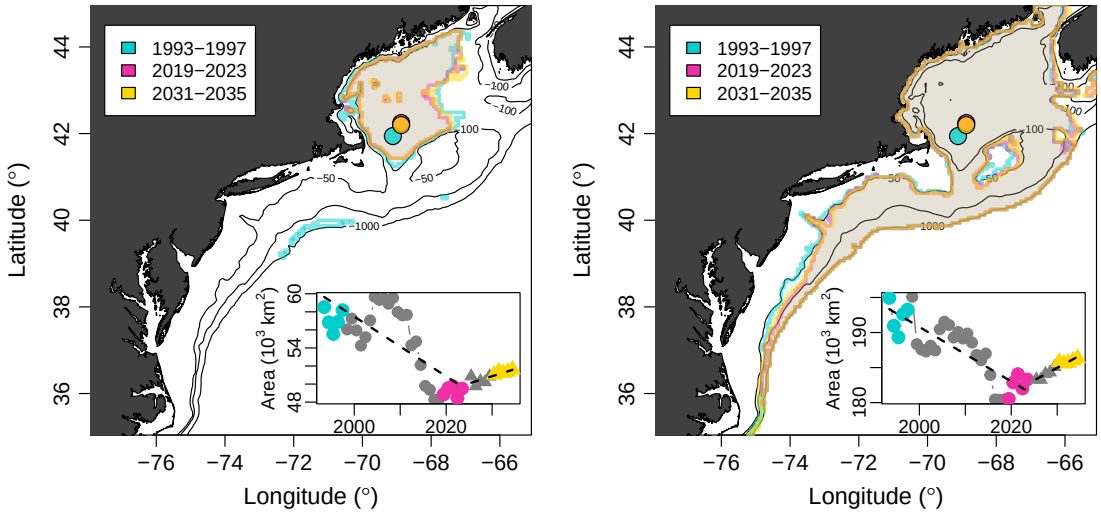


Figure 6: Distribution shifts on the Northeast Shelf for Witch flounder. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Witch flounder experienced significant core habitat and range loss between 1993 and 2023. Core habitat losses occurred on the continental shelf break and throughout the Gulf of Maine. Winter flounder lost range along the continental shelf, with the range contracting towards the shelfbreak and away from the coast in the Mid Atlantic Bight (Figure 6). The weighted centroid shifted 39 km northeast.

Under near-future ocean conditions, winter flounder core habitat and range is expected to expand. Witch flounder are predicted to gain habitat in the Gulf of Maine. Their range is expected to expand westward on the continental shelf towards the coast in the Gulf of Maine (Figure 6). The weighted centroid is predicted to shift 5 km south.

In the previous Assessment, witch flounder received a High Distributional Vulnerability Rank (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	57,399	192,655	4.99	6.65
2019-2023	39.0 km NE	49,226 (-8,172)	182,690 (-9,965)	4.36 (-0.62)	5.85 (-0.80)
2031-2035	5.1 km S	51,306 (+2,079)	185,858 (+3,169)	4.55 (+0.19)	6.17 (+0.32)

Directional Effect:

Experts predicted that witch flounder will be **negatively** impacted by climate change, with certainty equal to 95%. This is similar, but with higher certainty, to the 2016 Climate Vulnerability Assessment, which also predicted negative impacts from climate change on witch flounder (Hare et al. 2016). While there are few direct studies of the impacts of climate change on witch flounder, adults are known to prefer relatively cold temperatures between 4-11 C (Cargnelli et al. 1999). A recent study suggests that distributions are shifting southward (Mills et al. 2024).

Climate Effects on Abundance and Distribution

There have been few direct studies on the effect of climate change on witch flounder. Larval growth and survival has been linked to temperature in laboratory conditions (Bidwell and Howell 2001). Murawski (1993) included witch flounder in a list of species that exhibit low variation in seasonal temperature or depth of occurrence. Mills et al. (2024) found that witch flounder distribution has actually shifted significantly toward more southern latitudes.

Life History Synopsis

Witch flounder are a long-lived flatfish species inhabiting deep, cold waters in the Northwest Atlantic from Labrador to North Carolina (Cargnelli et al. 1999). They are slow-growing, reaching maturity at 4.5 years for males and 6 years for females in Georges Bank and the Gulf of Maine (Burnett, Ross, and Clark 1992). Witch flounder can live up to 30 years, reaching a maximum length of 78 cm (Cargnelli et al. 1999). Witch flounder spawn from March to November, with peak spawning occurring in summer. The general trend is for spawning to occur progressively later from south to north. In the Gulf of Maine-Georges Bank region, spawning occurs from April to November, and peaks from May to August (Martin and Drewry 1978) (Burnett, Ross, and Clark 1992). Powles and Kohler (1970) suggested that witch flounder occupy three largely discrete ecological niches, resulting in the different life stages being geographically separated from each other. The pelagic egg and larval stages are spent in the water column over deep water. Egg hatching occurs 7 to 8 days after spawning, with the yolk sac absorbed 10 days after hatching. Larvae are pelagic, and the larval stage lasts from 4-6 months to 1 year (Cargnelli et al. 1999). Eggs generally drift southward, and shelf currents may transport some larvae. Length may be the biggest cue for settlement (Cargnelli et al. 1999). The juvenile stage occurs in very deep water when metamorphosed fish settle to the bottom. They are found over mud, clay, silt, or muddy sand substrates (Cargnelli et al. 1999). Juveniles remain separated from the adult population, occupying deeper areas

until the onset of sexual maturity. Walsh (1987) suggests however that while the juveniles and adults are in different areas, the niche separation may not be true. Witch flounder are capable of inhabiting depths up to 1500 m, although the majority are found at 90-300 m (Martin and Drewry 1978) (Klein-MacPhee 2002). The main food items in the witch flounder diet are polychaetes and crustaceans, although mollusks and echinoderms are also important. Overall, polychaetes were by far the most important food item, accounting for greater than 70% of the diet (Maurer and Bowman 1975). However, there is a distinct ontogenetic shift in diet, with polychaetes increasing in importance and crustaceans decreasing in importance with age (Cargnelli et al. 1999). Adult witch flounder generally do not migrate, although they prefer to stay within temperatures of 4-11°C. Some subpopulations may move from shallower waters in warmer months to deeper overwintering waters. Migration was not observed in the Gulf of Maine (Cargnelli et al. 1999). Predators of witch flounder include monkfish, spiny dogfish, harp seals, Atlantic halibut, white hake, and some skates (Klein-MacPhee 2002). Witch flounder are managed as a single stock unit by the New England Fishery Management Council under the Northeast Multispecies (Groundfish) Fishery Management Plan. In the last Management Track Assessment, the stock status could not be determined due to a lack of biological reference points. The stock condition appears to be improving but is poor relative to historical levels (Northeast Fisheries Science Center (NEFSC) 2024).

Supplemental Scores

Exposure Timeseries

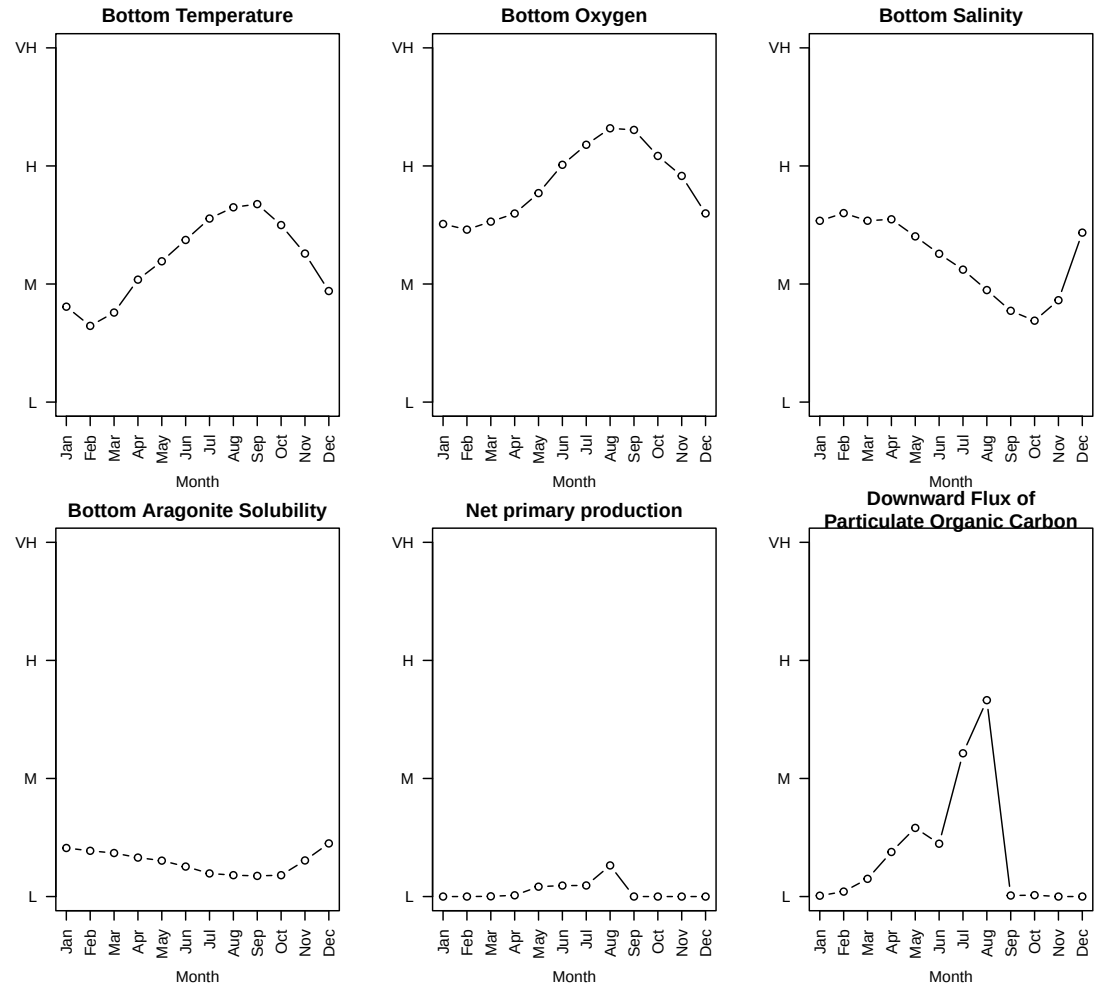


Figure 7: Monthly Timeseries of each Exposure Factor for Witch flounder. Exposure was calculated between 2014-2023 and 2025-2035

Exposure oscillated between Low and Moderate on average across months. Exposure was Moderate in January, March - April, and July - September (Figure 8). These patterns were driven by coincident peaks in bottom temperature, oxygen, and salinity as well as downward particulate organic carbon flux (Figure 7).

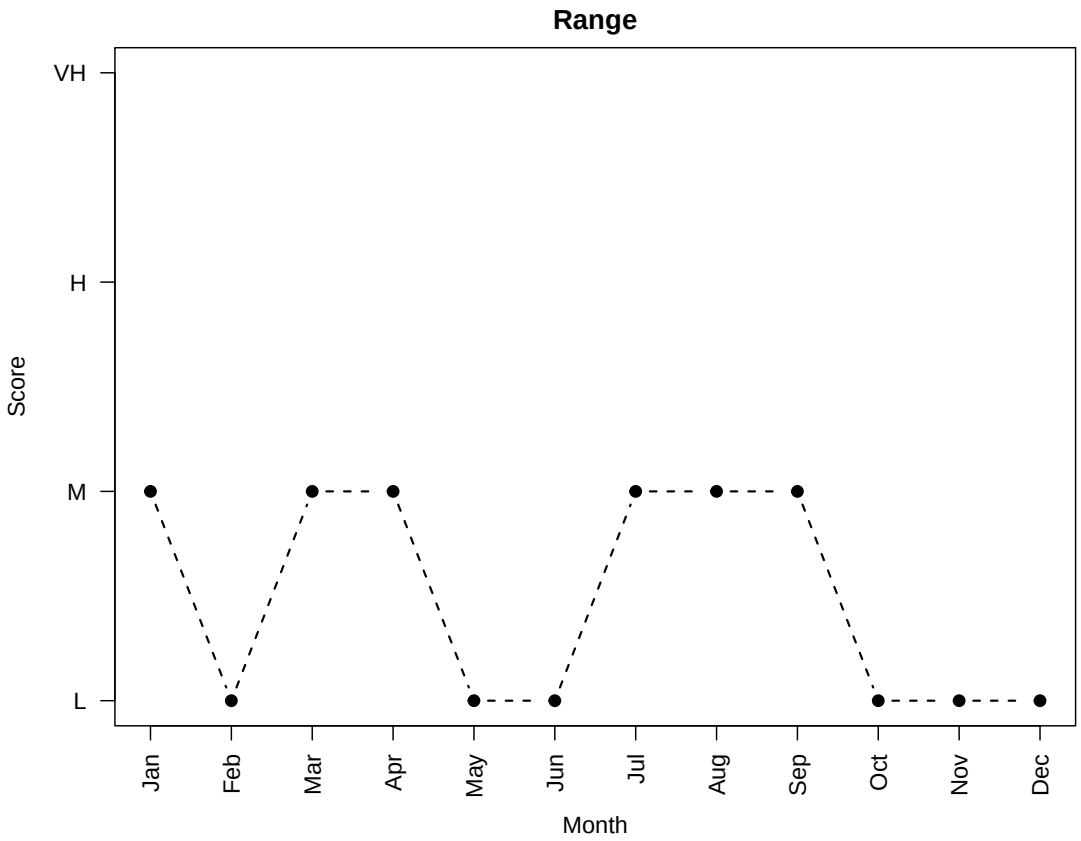


Figure 8: Timeseries of Total Exposure for Witch flounder

Exposure due to Significant Factors

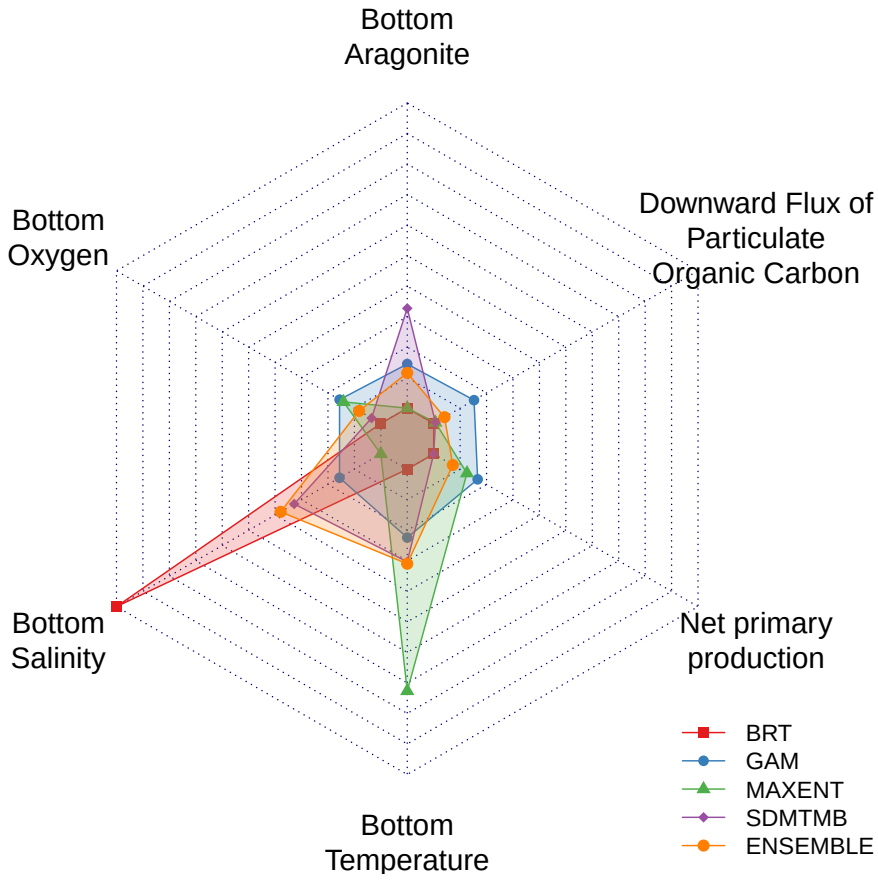


Figure 9: Dynamic Factors driving Witch flounder distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, salinity, and aragonite solubility were important dynamic drivers of witch flounder distributions (Figure 9). Witch flounder occupy cold, deep waters (Cargnelli et al. 1999). Diets are dominated by polychaetes but mollusks and echinoderms also play important roles (Maurer and Bowman 1975). When only these factors were used to calculate exposure, mean exposure decreased to Low (1.48 +/- 0.79) (Figure 11; Figure 10).

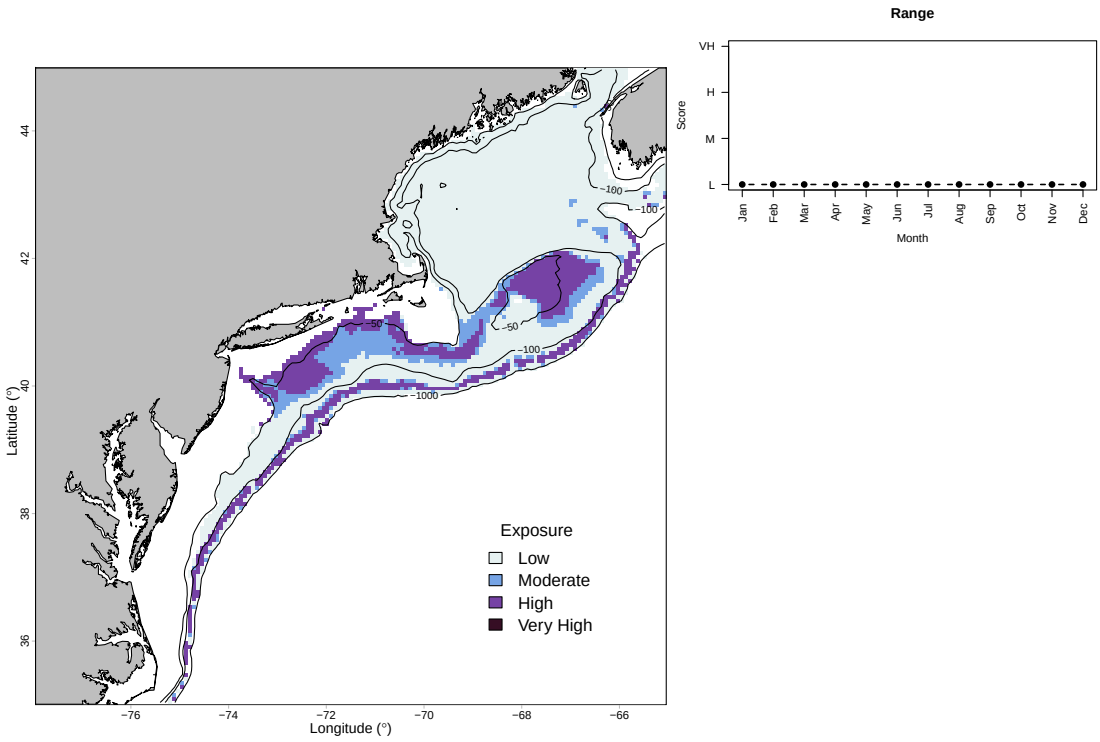
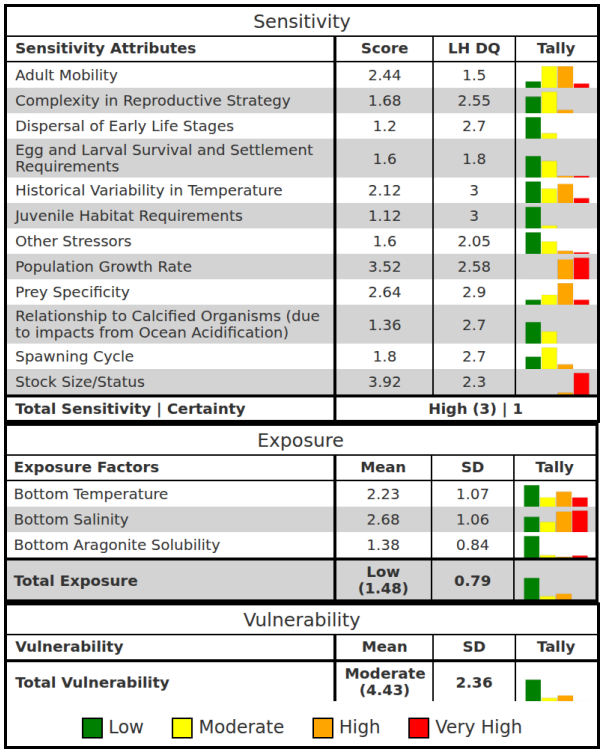


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Witch flounder distributions



Sensitivity			
Sensitivity Attributes	**Score**	**LH DQ**	**Tally**
Adult Mobility	2.44	1.5	
Complexity in Reproductive Strategy	1.68	2.55	
Dispersal of Early Life Stages	1.2	2.7	
Egg and Larval Survival and Settlement Requirements	1.6	1.8	
Historical Variability in Temperature	2.12	3	
Juvenile Habitat Requirements	1.12	3	
Other Stressors	1.6	2.05	
Population Growth Rate	3.52	2.58	
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Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.36	2.7	
Spawning Cycle	1.8	2.7	
Stock Size/Status	3.92	2.3	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure			
Exposure Factors	**Mean**	**SD**	**Tally**
Bottom Temperature	2.23	1.07	
Bottom Salinity	2.68	1.06	
Bottom Aragonite Solubility	1.38	0.84	
Total Exposure	**Low (1.48)**	**0.79**	

Vulnerability			
Vulnerability	**Mean**	**SD**	**Tally**
Total Vulnerability	**Moderate (4.43)**	**2.36**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Witch flounder, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

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